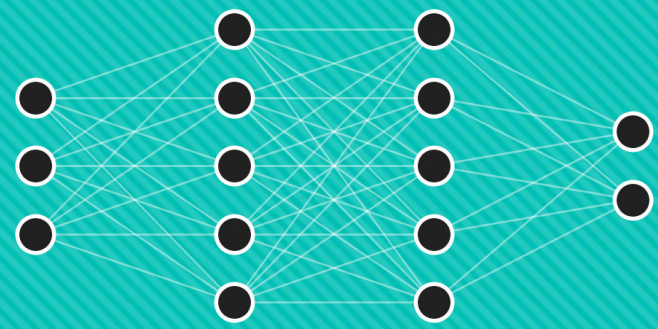
Salary Prediction Using Linear Regression
Group 11



Amogh Chavan :- Executive Lead

Puthineedi Vyshnavi :- Documentation Lead

Kshitiz Kumar :- Video Lead

Livteshwar Singh Sandhu :- PPT Lead

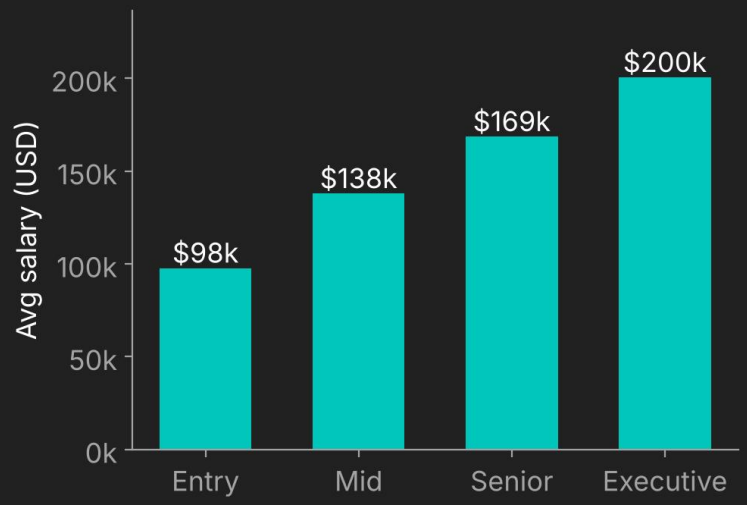
Git Hub Repo Link:-
https://github.com/2026esb1489-glitch/employee-salary-compensation-prediction

Problem and Data

- **Problem:** Predict an employee's annual salary (USD) with multiple linear regression.
- **Why:** Help organisations check and improve internal compensation equity.
- **Dataset:** "Latest Data Science Job Salaries 2020–2025" (Kaggle, by Saurabh Badole).
- **Size:** 93,597 rows × 11 columns; 46,637 rows remain after removing duplicates.
- **Target:** salary_in_usd
- **Features (7):** experience level, company size, employment type, job title, company location, work year, remote ratio.
- **Kaggle Dataset Link: https://www.kaggle.com/datasets/saurabhbadole/latest-data-science-job-salaries-2024/code**



Average salary rises with experience level



Approach



1 Data

Kaggle CSV with 93,597 rows and 11 columns

2 Clean

Drop duplicates, leaving 46,637 rows. Target is salary_in_usd

3 Preprocess

Fill missing values, encode the categories, scale the numbers

4 Model

Linear Regression, trained on 80% and tested on 20%

5 Output

Predicted salary, checked with R^2, MAE and MSE

- **Ordinal encoding:** experience_level (EN < MI < SE < EX) and company_size (S < M < L)
- **One-hot encoding:** employment_type, job_title and company_location, up to 10 categories each
- **Scaling:** work_year and remote_ratio, using StandardScaler
- **Libraries:** pandas, NumPy, scikit-learn, Matplotlib

Results

0.23
R^2 score

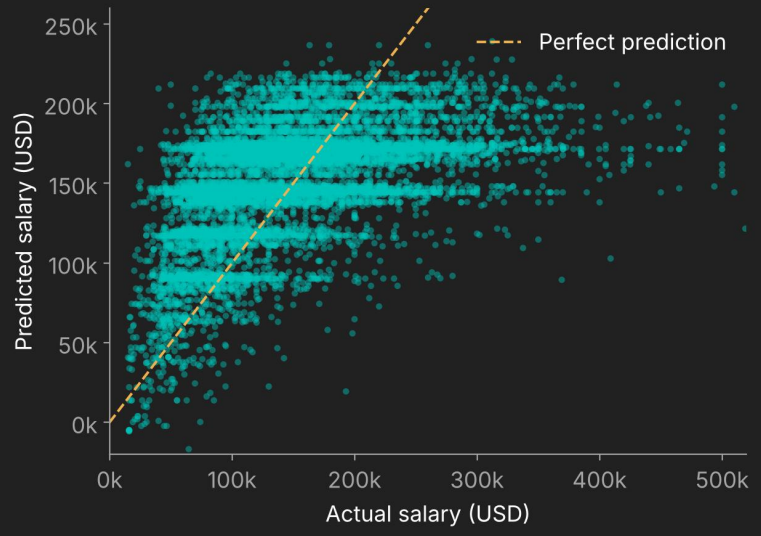
$49.5K
MAE (average error)

$67.3K
RMSE (√MSE)

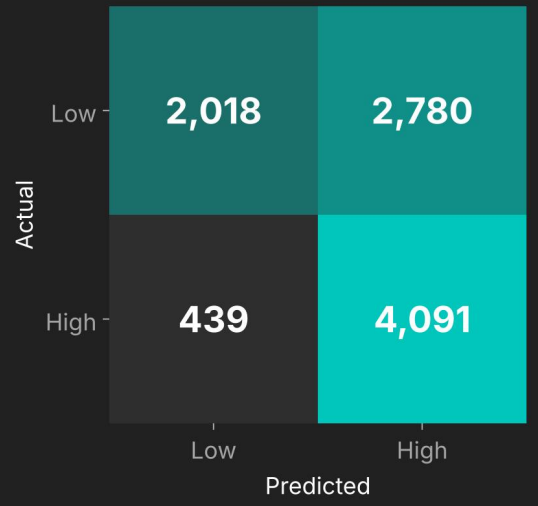
65.5%
Accuracy*

90.3%
Recall*

Actual vs predicted salary



High vs low salary*



What it shows

- The model gets the general trend right, but it under-predicts salaries above $300K.
- It catches most high earners (recall 0.90), but it also flags too many people as high (precision 0.60).
- R^2 of 0.23 means it explains only about a quarter of the salary variation.

*Salaries classified as high or low using the training-set median as the cut-off.

Learnings

What worked

- One-hot encoding let the model use job title and location.
- Putting everything in one scikit-learn pipeline kept the preprocessing tidy.
- Linear regression gave us a simple baseline we could explain.

What didn't

- We had no regional or cost-of-living data, so accuracy stayed low ($R^2 = 0.23$).
- Removing too many outliers cost us a lot of data.
- Next time we would try a non-linear model like Random Forest.

Who did what

- Amogh Chavan: Executive Lead
- Puthineedi Vyshnavi: Documentation Lead
- Kshitiz Kumar: Video Lead
- Livteshwar Singh Sandhu: PPT Lead

AI-use disclosure

- The PPT was made by ppt lead but cleaned up by Claude in features such as approach slide steps
- ChatGPT was used to assist with understanding the project and preparing initial practice code
- Claude helped write the train/test split, evaluation and plot cells; we ran and checked the output.